Evaluation and baseline equations

SWOTxAI

3 November 2026

1 Notation

One row is one radar grid cell c at one SWOT pass time t. The target is the 25-hour centred mean of the HFRNet surface current at that cell and time, $\mathbf{y} = (y_u, y_v)$. Inputs at the row:

$\mathbf{g} = (g_u, g_v)$	SWOT L3 geostrophic velocity at the centre cell,
$\mathbf{W} = (W_u, W_v)$	ERA5 10-m wind at the pass time,
$\mathbf{d} = (d_u, d_v)$	DUACS L4 geostrophic velocity interpolated to the row,
$\mathbf{x} \in \mathbb{R}^{72}$	the 3×3 stencil of eight SWOT, wind and MDT variables (81 columns on the calval orbit, where GOES SST is included),
$\tilde{\mathbf{x}}$	the stencil standardised by training-column mean and standard deviation, missing values set to zero after standardising.

$\bar{\mathbf{y}}$ is the mean of $\mathbf{y}$ over the training rows of the region; $\bar{\mathbf{y}}(c)$ is the mean over the training rows at cell c. Rows are split in time: training cycles first, a buffer of B days after the last training pass (30 days on the science orbit, 7 on calval), then the held-out passes.

2 Scoring

For one region and one component, with n held-out rows, prediction $\hat{y}_i$ and held-out mean $\langle y \rangle$,

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \langle y \rangle)^2}, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{1}$$

The sums run over every held-out cell and pass of the region pooled together, so the denominator contains the between-cell spread of the mean flow as well as the pass-to-pass variability. Confidence intervals are a block bootstrap over (cycle, pass): 1000 resamples of whole passes with replacement, 2.5 and 97.5 percentiles.

The skill score against the per-cell climatology (eq. 7) removes the between-cell credit:

$$S = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i \left(y_i - \bar{y}(c_i)\right)^2}, \tag{2}$$

which is zero for the climatology itself and positive only when a model knows something the map of the mean flow does not.

3 Baselines

Every fitted coefficient below is obtained by ordinary least squares on the training rows of the region, separately for u and for v (each component has its own coefficients of the same form), and applied unchanged to the held-out rows. Only the u form is written.

Fixed rules, no parameters.

$$\hat{u} = \bar{y}_u \qquad \text{regional training mean,} \tag{3}$$

$$\hat{u} = g_u \qquad \text{SWOT geostrophy, slope one,} \tag{4}$$

$$\hat{u} = d_u \qquad \text{DUACS geostrophy, slope one,} \tag{5}$$

$$\hat{u} = g_u + \epsilon\,(W_u \cos\theta + W_v \sin\theta) \qquad \text{Ekman rule, } \epsilon = 0.03,\ \theta = 45^\circ, \tag{6}$$

$$\hat{v} = g_v + \epsilon\,(-W_u \sin\theta + W_v \cos\theta) \qquad \text{(wind rotated } \theta \text{ to the right).}$$

Climatology (the map).

$$\hat{u} = \bar{y}_u(c), \qquad \bar{y}_u(c) = \frac{1}{n_c} \sum_{i \in \text{train}(c)} y_{u,i}, \tag{7}$$

the training-period mean at the row's own cell; a held-out cell with no training rows falls back to $\bar{y}_u$.

Linear rungs. With (a, b, c, d, e) fitted per component:

$$\hat{u} = a + b\,W_u + e\,W_v \qquad \text{wind only,} \tag{8}$$

$$\hat{u} = \bar{y}_u(c) + a + b\,W_u + e\,W_v \qquad \text{climatology + wind, fitted to } y_u - \bar{y}_u(c), \tag{9}$$

$$\hat{u} = g_u + a + b\,W_u + e\,W_v \qquad \text{SWOT (slope one) + wind, fitted to } y_u - g_u, \tag{10}$$

$$\hat{u} = a + b\,g_u \qquad \text{SWOT calibrated,} \tag{11}$$

$$\hat{u} = a + b\,g_u + c\,g_v + d\,W_u + e\,W_v \qquad \text{SWOT calibrated + wind,} \tag{12}$$

$$\hat{u} = a + b\,d_u \qquad \text{DUACS calibrated,} \tag{13}$$

$$\hat{u} = a + b\,d_u + c\,d_v + d\,W_u + e\,W_v \qquad \text{DUACS calibrated + wind.} \tag{14}$$

Per-cell linear (partially pooled). The five-parameter model of eq. 12 fitted at each cell on standardised inputs and shrunk toward the regional fit. With A_c the $n_c \times 5$ design matrix of cell c (a column of ones and the four standardised inputs), $\mathbf{y}_c$ its training targets, and $\boldsymbol{\beta}_g$ the regional least-squares coefficients,

$$\boldsymbol{\beta}_c = \left(A_c^\top A_c + \lambda I\right)^{-1}\left(A_c^\top \mathbf{y}_c + \lambda\,\boldsymbol{\beta}_g\right), \qquad \lambda = 5, \tag{15}$$

so the prior is worth five training rows; cells with fewer than five complete training rows use $\boldsymbol{\beta}_g$. With about 170 rows per cell on the calval orbit the local fit dominates; with about ten on the science orbit the prior does.

Ridge on the full stencil.

$$\hat{u} = a + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}, \qquad \boldsymbol{\beta} = \arg\min_{\boldsymbol{\beta}} \sum_{i \in \text{train}} \left(y_{u,i} - a - \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i\right)^2 + \alpha\,\|\boldsymbol{\beta}\|^2, \qquad \alpha = 1. \tag{16}$$

This is the linear ceiling on the champion's own inputs.

AlphaEvolve champion.

$$(\hat{u}, \hat{v}) = f_\theta(\tilde{\mathbf{x}}), \tag{17}$$

a multilayer perceptron with joint (u, v) output, trained by minimising the squared error masked to finite targets, with early stopping on a validation split; the architecture and training schedule are the candidate program selected by the evolutionary search (`candidate.py` of the run).

4 Alternative truth tiers

The same predictions are also scored against three other versions of the radar current built from the hourly record $u(t)$ at each cell: the raw hourly value; and two zero-phase Butterworth low-passes of order 2, applied forward and backward (`filtfilt`), with cutoff periods of 18 and 48 hours. The headline 25-hour target is the centred boxcar

$$\bar{u}_{25}(t) = \frac{1}{25} \sum_{k=-12}^{12} u(t + k\,\mathrm{h}), \tag{18}$$

whose power response $|H(f)|^2$ has its first null at $f = 1/25\ \mathrm{h}^{-1} = 0.96$ cycles per day. Tiers are scored on the rows where every tier is finite.